End-Term Sample Question Paper - ANSWER KEY

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set D - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

For the instructor. Every numerical result in this key was recomputed and verified before printing; the marks in brackets give the part-wise marking scheme.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. State the two steps of a proof by mathematical induction, and explain in one sentence why a proof that omits the basis step is worthless. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

Basis: prove the statement for the first value. **Inductive step:** assume it holds for an arbitrary k and prove it for k + 1. (1 mark)

Without the basis the chain has nothing to start from, so a **false** statement can satisfy the inductive step alone and still be 'proved'. (1 mark)

Q2. Given f(0,0) = 10, f(1,0) = 30, f(0,1) = 50 and f(1,1) = 70, compute the intensity at (0.25, 0.75) by bilinear interpolation, and state the nearest-neighbour value. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

With a = 0.25, b = 0.75 the weights are 0.1875, 0.0625, 0.5625, 0.1875 (summing to 1), giving 1.875 + 1.875 + 28.125 + 13.125 = **45.0**. (1.5 marks)

Nearest neighbour: (0.25, 0.75) → (0, 1), so the value is **50**. (0.5 marks)

Q3. Given P(A) = 0.7, P(B) = 0.5 and P(A ∩ B) = 0.35, compute P(A | B) and state whether A and B are independent, with the reason. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

P(A | B) = 0.35 / 0.5 = **0.7**. (1 mark)

P(A)P(B) = 0.7 × 0.5 = 0.35 = P(A ∩ B), and P(A | B) = P(A), so A and B **are independent**. (1 mark)

Q4. State the condition a directed graph must satisfy for a topological ordering to exist, and give a one-line reason. **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

The graph must be a **directed acyclic graph (DAG)**. (1 mark)

A cycle would require every vertex on it to appear before itself in the ordering, which is impossible. (1 mark)

Q5. State the worst-case running time of merge sort and of quick sort, and state the condition that triggers quick sort's worst case. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

Merge sort: **Θ(n log n)** in every case. Quick sort: **Θ(n²)** in the worst case. (1 mark)

Quick sort degrades when the pivot repeatedly produces maximally unbalanced partitions of sizes 0 and n − 1 — for example on already-sorted input when the first element is taken as the pivot. (1 mark)

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. Six activities are available with (start, finish) times (1,4), (3,5), (0,6), (5,7), (8,11) and (8,12). Apply the greedy activity-selection algorithm: state the sorting key, list the activities chosen, and state how many are selected. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Sort by **finish time**, then repeatedly take the next activity whose start is at least the previous finish. (1.5 marks)

Order: (1,4), (3,5), (0,6), (5,7), (8,11), (8,12). (0.5 marks)

Take **(1,4)**; reject (3,5) and (0,6) as they start before 4; take **(5,7)**; take **(8,11)**; reject (8,12), which starts before 11. (2 marks)

3 activities are selected: (1,4), (5,7), (8,11). (1 mark)

Q7. Simplify F(A, B, C) = Σm(1, 3, 5, 7) using a three-variable Karnaugh map. Write the minterms in binary, state the single group that covers them, give the minimal sum-of-products expression, and state the literal saving. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

Minterms: m_1 = 001, m_3 = 011, m_5 = 101, m_7 = 111. (1.5 marks)

In all four, C = 1 while A and B each take both values, so both cancel: the four cells form **one group of four**. (1.5 marks)

Minimal SOP: **F = C**. (1 mark)

Canonical form: 4 minterms × 3 literals = 12 literals; minimal form: 1 literal; saving = **11 literals**. (1 mark)

Q8. Explain what the correlation coefficient measures and state its range; state what R^2 represents; and explain in two sentences why a strong correlation does not establish causation. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

The correlation coefficient r measures the **strength and direction of a linear relationship**, with $-1 \le r \le 1$. (1.5 marks)

$R^2 = r^2$ is the **proportion of the variance in the response explained by the model**. (1.5 marks)

A correlation is consistent with X causing Y, Y causing X, or a third variable driving both (a confounder); it is also vulnerable to coincidence in small samples. Establishing causation needs an intervention or a design that rules those alternatives out. (2 marks)

Q9. State the Master theorem's three cases, and give one recurrence to which it does **not** apply, explaining why. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

For T(n) = aT(n/b) + f(n) with critical exponent n^(log_b a): **Case 1** f polynomially smaller → Θ(n^(log_b a)); **Case 2** f = Θ of it → Θ(n^(log_b a) log n); **Case 3** f polynomially larger and regular → Θ(f(n)). (3 marks)

T(n) = 2T(n/2) + n/log n does **not** fit: f(n) is smaller than n^(log_b a) = n but **not by a polynomial factor** n^ε, so it falls in the gap between Cases 1 and 2. (1.5 marks)

T(n) = T(n−1) + 1 is also outside its scope, since it does not divide the input by a constant factor. (0.5 marks)

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Linear Discriminant Analysis. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) State what Fisher's criterion maximises, and explain why LDA uses the within-class scatter rather than simply joining the two class means. *[5]*

(b) For C_1 = {(1,2), (3,2), (2,5)} and C_2 = {(6,4), (8,4), (7,7)}: compute both means, S_1, S_2 and S_W, the direction $w = S_W^{-1}(m_2 - m_1)$, and the threshold midway between the projected means. *[5]*

(a) Fisher's criterion maximises the ratio of **between-class** separation to **within-class** scatter, $(w^T(m_2-m_1))^2 / w^T S_W w$. (2.5 marks)

Joining the means ignores how **spread** each class is along a direction. A direction in which the means are far apart is useless if both classes are also very wide along it; dividing by S_W down-weights exactly those directions. (2.5 marks)

(b) m_1 = **(2, 3)**, m_2 = **(7, 5)**. (1 mark)

C_1 deviations (−1,−1), (1,−1), (0,2): $\Sigma d_x^2 = 2$, $\Sigma d_y^2 = 6$, $\Sigma d_x d_y = 0$, so S_1 = [[2, 0], [0, 6]]. C_2 has the identical pattern, so $S_2 = S_1$ and **S_W = [[4, 0], [0, 12]]**. (1.5 marks)

$m_2 - m_1$ = (5, 2), so w = [[1/4, 0], [0, 1/12]](5, 2)^T = **(5/4, 1/6)**, proportional to **(15, 2)**. (1.5 marks)

With w = (15, 2): C_1 projects to 19, 49, 40 (mean **36**); C_2 to 98, 128, 119 (mean **115**); threshold = (36 + 115)/2 = **75.5**. Every C_1 value is below it and every C_2 value above, so the classes separate cleanly. Note w weights x far more than y, because the within-class scatter is six times larger in y. (0.5 marks)

Q11. Network flow. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) State the max-flow min-cut theorem, explain what a residual graph is, and describe how the minimum cut is read off once the algorithm stops. *[5]*

(b) Find the maximum flow for capacities s→a = 12, s→b = 6, a→b = 3, a→t = 9, b→t = 8. Give each augmenting path and bottleneck, the maximum flow, the reachable set S, and the cut capacity. *[5]*

(a) The value of a maximum flow equals the capacity of a minimum cut. Every cut bounds the flow from above, and the bound from the smallest cut is achieved exactly. (2 marks)

The **residual graph** records remaining capacity: a forward edge of capacity c − f, plus a backward edge of capacity f allowing flow to be cancelled. (1.5 marks)

When no augmenting path remains, let **S** be the vertices still reachable from s in the residual graph and T the rest; the edges from S to T are all saturated and form a **minimum cut**. (1.5 marks)

(b) s→a→t, bottleneck min(12, 9) = **9**; s→b→t, bottleneck min(6, 8) = **6**; s→a→b→t, residuals 3, 3, 2, bottleneck **2**. (2 marks)

Maximum flow = 9 + 6 + 2 = **17** (check at the source: 11 + 6 = 17). (1 mark)

Residual s→a = 12 − 11 = 1 > 0, so a is reachable; a→b = 3 − 2 = 1 > 0, so **b is reachable too**; b→t and a→t are both saturated, so t is not. **S = {s, a, b}**, T = {t}. (1.5 marks)

Cut capacity = c(a,t) + c(b,t) = 9 + 8 = **17**, equal to the maximum flow as the theorem requires. (0.5 marks)

Q12. Genetic algorithms. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) Describe the encoding, fitness, selection, crossover and mutation operators, and explain the role of elitism. *[5]*

(b) A GA maximises $f(x) = x^2$ over 5-bit chromosomes starting from 01011, 11100, 00110, 10001. Decode each, compute the fitness, the total and average, and the roulette-wheel selection probabilities. *[5]*

(a) Encoding: represent a candidate as a chromosome (here a 5-bit string). **Fitness:** score each candidate by the objective. **Selection:** choose parents with probability rising with fitness (roulette wheel, tournament). (3 marks)

Crossover: combine two parents at a cut point to make offspring. **Mutation:** flip bits with small probability, which preserves diversity and stops premature convergence. **Elitism:** carry the best individual unchanged into the next generation, so the best fitness can never decrease. (2 marks)

(b) 01011 = **11**, 11100 = **28**, 00110 = **6**, 10001 = **17**. (2 marks)

Fitness: 121, 784, 36, 289. Total = **1230**, average = **307.5**. (2 marks)

$p_i = f_i / \Sigma f$: **0.0984, 0.6374, 0.0293, 0.2350** (summing to 1). Chromosome 2 takes nearly two-thirds of the wheel, which is how selection pressure concentrates the search — and why mutation is needed to keep the population from collapsing onto it. (1 mark)

----- End of Answer Key -----